

Rover Evidence Lab

I can carry out a comparative fair test, use repeated distance results for a conclusion and improve the method.

TEST

use evidence

RECORD

use evidence

CONCLUDE

use evidence

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What success looks like

- I can complete two conditions with three repeats each.
- I can record every distance and identify the pattern.
- I can write or communicate a conclusion, one improvement and one model limitation.

Retrieve and reconnect

- State the surface-test independent and dependent variables.
- Name two controls from the plan.
- Explain why the rover is released without a push.
- Predict which surface gives the greater distance and why.

Smooth-surface results are 48, 51 and 49 cm. What should happen to the 51 cm result?

Think first. Point, say, sign or write your choice. Give one reason.

- A. Keep it with the other valid repeats
- B. Delete it because it is not identical
- C. Keep only 51 because it is the furthest

Look closely · what is the evidence?

PREDICT · commit before testing

TEST · repeat the same method

RECORD · pattern, conclusion, improvement

Look closely · what is the evidence?

"I predict the rover goes farther on smooth, because..."

a prediction is held, not marked

Prediction

SMOOTH

48 cm · 51 cm · 49 cm
mean = 49 cm

FABRIC

27 cm · 30 cm · 28 cm
mean = 28 cm

Results

Pattern: every fabric run is shorter.

Conclusion: more friction on fabric → shorter distance.

Improvement: a release gate so every start is identical.

Conclusion

Words we will use

- repeat — same method again
- conclusion — what the results support
- limitation — what the test cannot show

Watch the model

- Check the safe lane, low ramp, marked start and same rover. Commit a prediction before releasing anything.
- Release without a push, measure from the same line and record immediately. Repeat three times before changing condition.
- Keep every valid result. If the method fails, preserve the first note, repair the setup and repeat the run.

Which recording choice gives the strongest evidence?

- A. Record all three valid results for each condition using the same measuring rule.
 - B. Keep only the furthest result from each condition.
 - C. Change a result when it does not match the prediction.
- Commit to an answer. Show the evidence you used. Listen, then revise if needed.

Build the explanation

- New investigation evidence: smooth 48, 51, 49 cm; fabric 27, 29, 28 cm. Every smooth result is greater.
- Typical distance is about 49 cm on smooth and 28 cm on fabric. The observed intervals 48–51 cm and 27–29 cm do not overlap; the numerical ranges are 3 cm and 2 cm.
- Conclusion: the rover travelled farther on smooth surface because lower friction opposed its motion less. State that the classroom model is simplified.

Find and repair the misconception

Keep the furthest result and discard the rest because it shows the rover's best performance.

A. Keep the claim: only the best result matters.

B. Repair it: keep every valid repeat so the pattern and variation remain visible.

C. Change it: write one average and erase all raw results.

Rover Results Lab

Place each evidence card under Smooth, Fabric or Conclusion. Freeze the pattern before choosing an improvement.

- Predict
- Record three repeats
- Compare the pattern
- Conclude and improve

Use the numbered cards in your pupil pack. Take turns to choose, justify and check. You may direct a partner to move a card.

Choose a group · defend your placement

- SMOOTH
- FABRIC
- CONCLUSION / LIMIT

State the pattern, one evidence-linked conclusion, one method improvement and one model limitation.

Your independent mission

Carry out or direct the fair rover test, record two conditions with three repeats, then give a conclusion, improvement and limitation.

Record: Prediction, all six raw results, pattern statement, evidence-linked conclusion, one improvement and one model limitation.

Choose your support and challenge

- Supported: Direct each run, match the six values to conditions and complete an oral or symbol-supported conclusion.
- Standard: Complete three safe repeats per condition, record all results and write a conclusion plus one improvement.
- Stretch: Compare typical values, calculate numerical ranges, explain variation and evaluate transfer to a real rover.

Show what you know

Use one repeated result pattern to explain the effect of surface on the rover.

Point to, read or explain the evidence you made. Choose one next step after the adult has listened.